

Figure 2. An example of a Data Science Life Cycle.

The light-yellow layer is the Application/Domain Science level. This level portrays the steps of a research project described at the domain level. The green layer beneath is the Infrastructure level, describing the computational infrastructure that enables the Application level. The blue arrow underlying both the application and infrastructure levels is the System level, describing system elements upon which the data science discovery process depends. Similarly, the overarching grey arrow is the meta-scientific level "The Science of Data Science." The arrows are intended to depict the life cycle: that the output of data science discovery (the findings and the artifacts that enable reproducibility such as code, data, and workflow and computational environment information for example) are published for verification and reuse by others. The figure provides a way to consider tool use explicitly for each discovery step. Some tools may support more than one discovery step, for example, notebooks, and some application level steps may use more than one tool. Different steps in the discovery pipeline, whether at the domain, infrastructure, system, or science of data science level, may be carried out by different people.

